

# CardCapsule — Capture Flow (v1.0)

## 1) Product goals & constraints

- **Primary outcome:** fast, low-friction capture that defaults to the common case (*Front + Inside*) while gracefully supporting **Back**, **multi-page insides**, **letters**, and **single-panel cards**.
- **Rules from brief:**
  - User must capture  $\geq 1$  **image** to proceed to Review.
  - Stay in **Capture** until the user taps **Continue** → **Review**.
  - User can **retake/edit/delete** any image during Capture.
- **Cancel:**
  - If nothing captured → instantly exit.
  - If  $\geq 1$  image captured → confirm destructive: *"This card will be deleted. This can't be undone."*
- Starting a new card should **open the camera immediately**.

## 2) Mental model: "Capture Stack"

A flexible, labeled queue of shots the user intends to take. We pre-seed the stack with **Front** and **Inside** (majority case). Users can add **Back** and **Extra page(s)** on demand, or remove anything. The stack gives guidance without forcing a rigid flow.

**Stack defaults (new card):** [Front] [Inside] [+ Back] [+ Extra]

- Tapping **Front/Inside** opens live camera if not captured; shows the existing photo if captured.
- Chips show status: ○ pending, ● captured, ✎ edited.
- **+ Back** adds a Back slot; **+ Extra** adds Inside 2, then Inside 3, etc.

## 3) User flow (high level)

Add New Card → (Camera opens immediately)

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Capture Screen (Stack seeded: Front, Inside)

- └ User captures Front
  - └ Auto-prompt: "Flip to Inside?" [Capture] [Skip]
- └ User captures Inside (optional)
  - └ Suggest: "Add Back?" [Add Back] "Add another Inside?" [+ Extra]
- └ User may add Back / Extra pages / switch to Letter mode
- └ Thumbnail tray shows all captured; tap to edit/retake/delete
  - └ [Continue → Review] (enabled once  $\geq 1$  captured)

Cancel (top-left)

- └ 0 captured → exit immediately
- └  $\geq 1$  captured → Destructive confirm → exit & delete temp

## 4) Screen specs — Capture

### 4.1 Top bar

- **Left:** Cancel
- **Center:** New Card
- **Right:** Continue → (disabled until  $\geq 1$  capture)

### 4.2 Viewfinder

- Full-bleed camera preview with **card outline** guidance (rounded rectangle with corner brackets).
- **Hints** (fade in/out): “Fill the frame”, “Avoid glare”, “Hold steady—auto capture...”.
- **Auto-capture** when edges and focus are stable; always allow manual shutter.

### 4.3 Controls

**Primary row (chips = Capture Stack):** [●/○ Front] [●/○ Inside] [ + Back ] [ + Extra ]  
[... More]

- **States:** ○ Pending, ● Captured, 🖋 Edited
- **Interactions:**
  - Tap pending chip → focuses camera on that slot.
  - Tap captured chip → opens **Image Detail** (Retake/Edit/Delete).
  - + **Extra** adds Inside 2, Inside 3, etc. (label auto-increments).
  - ... **More** opens a sheet for rare variants (see 6. Variants).

**Bottom bar:** - Left: Flash (Auto / On / Off) - Center: **Shutter** (big circle). Long-press → burst (picks best). - Right: Gallery (pick from camera roll)

**Thumbnail tray (floating, bottom-right)** - Stack of tiny thumbs (1–4 visible). Tap opens Image Detail for that shot. - Shows count badge if >4.

### 4.4 Image Detail (per shot)

- Full preview with:
- **Retake** (reopen camera with same slot selected)
- **Edit** (crop/rotate/deskew/contrast; auto-deskew suggested)
- **Replace** (choose from gallery)
- **Delete** (confirm; removes the slot or marks it pending if default slot)
- **Re-label** (optional) if AI mislabels; default shows the slot label (“Front”, “Inside 2”).

### 4.5 Micro-interactions

- **Haptics:** light tick on capture; warning vibration on blur/glare.
- **Toasts:** “Saved Front”, “Inside replaced”, “Back deleted”.
- **Smart prompts** after a capture:
  - After **Front**: “Flip open to capture Inside?”
  - After **Inside**: “Add Back or another Inside page?”

## 5) State logic (React-friendly)

```
// minimal shape
{
  mode: 'card' | 'letter' | 'unknown',
  shots: [
    { id: 'front', label: 'Front', type: 'front', status: 'pending|captured|
edited', uri?: string, edits?: {crop, rotate, deskew, contrast},
aiLabel?: 'front' | 'inside' | 'back' },
    { id: 'inside-1', label: 'Inside', type: 'inside', status: 'pending|captured|
edited', uri?: string },
    // dynamically add: {id: 'back', type: 'back'} and {id: 'inside-2',
'inside-3', ...}
  ],
  capturedCount: number,
  canContinue: boolean, // capturedCount >= 1
  isDirty: boolean, // capturedCount >= 1
}
```

**Actions:** `selectSlot(id)`, `capture(id, uri)`, `retake(id)`, `edit(id, edits)`, `delete(id)`, `addExtraInside()`, `addBack()`, `relabel(id, label|type)`.

## 6) Variants & edge cases

- **Single-panel card** (front only): system detects no fold → suggest “Looks like a single-panel card. You can continue now or add Back.”
- **Letter** (single page): quick toggle in ... More or auto-detect aspect → stack becomes `[Letter] [+ Back]`.
- **Back with message**: after Inside capture, nudge: “Is there a note on the back?” → `[Add Back]`.
- **Multi-page insides**: every tap on + **Extra** creates `Inside N`.
- **Gallery import**: if user imports multiple, open **Labeling sheet** to map each image to Front/Inside/Back/Inside N.
- **Offline**: persist locally until Review; show a small offline badge; edits all local.

## 7) Error & quality handling

- **Glare/blur/low light** detection → inline hint + allow capture; don't block.
- **Auto-deskew & crop** after capture; user can adjust in Edit.
- **Orientation** locked to correct upright using EXIF + edge detection.

## 8) Cancel & destructive confirm

- **Cancel (top-left)**:
- If no images: exit to previous screen.



## 12.2 After capturing Front

Top toast: ✓ Front saved [Flip to Inside? Capture | Skip]  
Chips: [● Front] [○ Inside] [ + Back ] [ + Extra ] [ ... ]  
Continue ▶ becomes enabled  
Thumb tray shows tiny Front thumbnail

## 12.3 With multiple captures

Chips: [● Front] [● Inside] [ + Back ] [ + Extra ] [ ... ]  
Thumb tray: [Front] [Inside] [Back?] ... (tap → Retake/Edit)  
Continue ▶ (enabled)

## 12.4 Image Detail (sheet)

[Front]  
| full preview |  
[Retake] [Edit] [Replace] (Delete)

## 13) Review handoff (for context)

- On **Continue** → **Review**, pass ordered shots with labels, edits applied.
- Review screen can perform OCR/transcription, let user reorder/rename, and confirm.

## 14) Future enhancements

- **Auto-label** side using vision model; let users override.
- **Smart sequence**: if Front found → auto highlight Inside chip.
- **Glare guard**: if severe glare, suggest angle change with micro-animation.
- **Batch mode**: capture several cards back-to-back, review at the end.

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**Why this works:** Defaults to *Front* + *Inside* (speed) while the Stack keeps the flow open-ended (flexibility). Users see progress, can fix mistakes inline, and can bail safely. It respects edge cases without cluttering the main path.